

rDNA Technology & Genome Editing



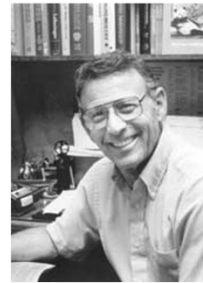
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Genotype	VS	Phenotype
		
Genotypes are the genetic make-up of an individual.		Phenotypes are the set of observable characteristics of an individual resulting from the interaction of its genotype with the environment.

Genetic Engineering.....

- A molecular genetic technique used for the direct manipulation, alteration or modification of genes or genome of organisms in order to manipulate the phenotypes is called genetic engineering
- In this technique, a recombinant DNA is constructed and inserted into the host genome using a vector. Or we can delete some mutant sequences from a genome. The first recombinant DNA was constructed by Paul Berg in 1972.

In conjunction with his studies of the tumor virus SV40, in 1972, Paul Berg succeeded in inserting DNA from a bacterium into the virus' DNA. He thereby created the first DNA molecule made of parts from different organisms. This type of molecule became known as "hybrid DNA" or "recombinant DNA".



Paul Berg

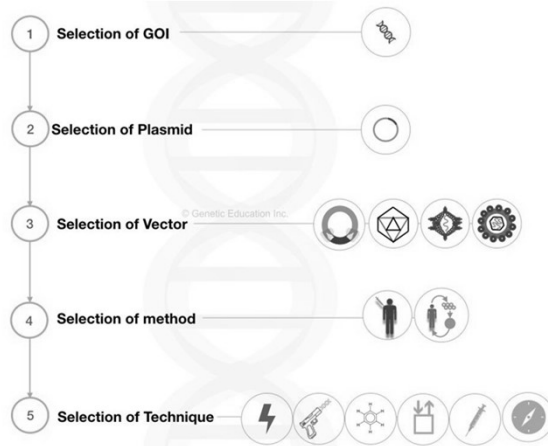
Genetic Engineering techniques....

- **Recombinant DNA-** A recombinant DNA technology is a type of genetic engineering technology in which an artificial DNA molecule is constructed by ligating two different DNAs using physical methods. For that, the gene of interest is inserted into the plasmid vector and used for gene transfer experiments.
- **Gene delivering-** Gene delivering technique is employed for the insertion of a gene of interest into the host genome.
- **Electroporation,** sonication and viral vector-mediated gene transfer, liposome-mediated gene transfer, transposon-mediated gene transfer are some of the methods used for that.
- **Gene editing-** A gene-editing technique is used to edit the genome in which an undesired DNA sequence is removed or a new gene can be inserted into the host genome. CRISPR-CAS9, TALEN and ZFN are some known gene-editing tools used in gene therapy experiments.



CRISPR: Clustered regularly interspaced short palindromic repeats; CAS-9: CRISPR associated protein 9; TALEN: Transcription activator-like effector nucleases ; ZFN: Zinc-finger nucleases

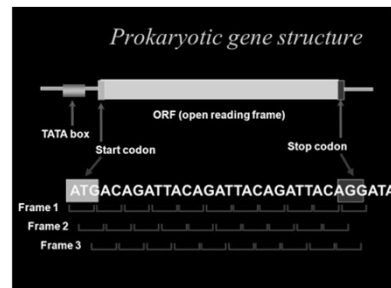
Process



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Selecting and isolating the candidate gene

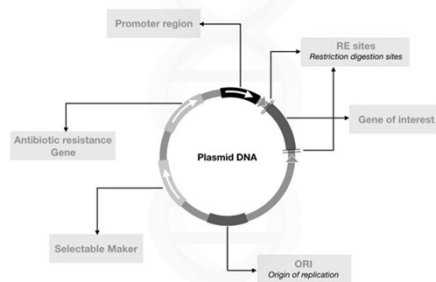
- **Special character:** The gene must contain a sequence of DNA that we want to study and for that, a gene has some special characteristics. A candidate gene should have high GC content and a lower repetitive DNA sequence.
- **Size:** The gene of interest must not be too long- only a few kb genes can be successfully inserted. Longer the gene higher the chance of failure. The candidate gene must have a start and stop codon in it.
- **Isolation:** Now, the gene of interest can be isolated from the rest of the DNA using either **Restriction Digestion** or **PCR**.
- **Artificial Synthesis:** If the gene of interest is well studied, previously, then the information of a gene is accessible in the genetic library and we can use it for the artificial synthesis of a gene of our interest. (using the genetic library information, the gene can also be artificially synthesized)



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Selection and construction of plasmid:

- The plasmid DNA is a circular, double-stranded cytoplasmic DNA of the bacteria that replicate independently.
- Selecting plasmid for the genetic engineering experiment is one of the crucial steps in the entire experiment. Before selecting the plasmid, we must understand why the plasmid is used in the gene transfer experiments.
- Scientists are using it as a vehicle for transferring the gene of interest to the target location in the genome. It can efficiently transfer the gene at the target location. The structure of plasmid is explained in the figure below.



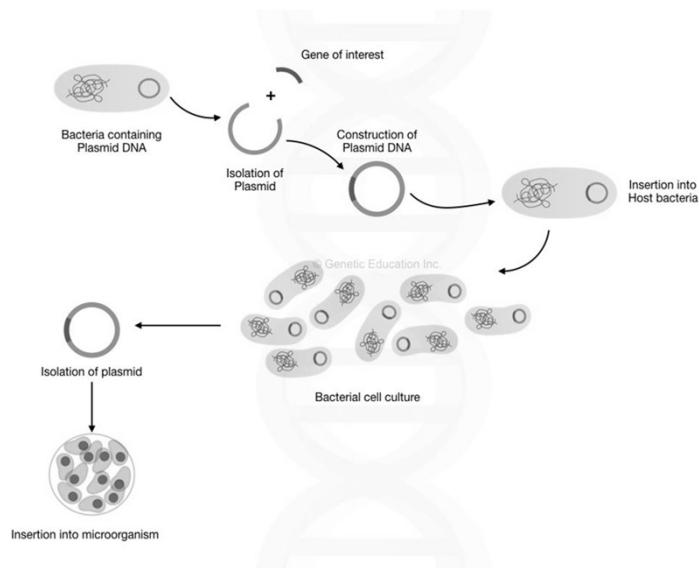
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Preparation of plasmid

Select the plasmid which suits your experiment.

- The plasmid must have the origin of replication, promoter region, antibiotic resistance gene and other important sequences. Using the restriction digestion method, an insertion site is introduced in the plasmid at which our gene of interest is ligated.
- Utilizing the T4 DNA ligase like power sealer, the DNA of our interest is inserted and ligated in the plasmid. Along with the plasmid, a selectable marker is also introduced in the plasmid DNA to identify the recombinant DNA.
- In addition to this, a promoter region and terminator sequences are also included in the plasmid for the effective expression of a gene of our interest. A plasmid with our gene of interest and some other important sequences is now referred to as a recombinant DNA molecule. Now our recombinant DNA is ready for expression.
- If we are performing gene cloning then the plasmid is inserted in the bacterial host, for that generally *E.coli* are commonly used. Once the bacteria starts dividing, our recombinant plasmid DNA is also replicated along with it.
- Now we have the multiple copies of our plasmid DNA which are extracted using the plasmid DNA extraction kit and used for the transformation experiments.

Preparation of plasmid



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Transformation into the host genome

- Various methods for recombinant DNA insertion is used for various cell types because a single method can't used for all cell types.
- **Using stress-** bacteria easily uptake the plasmid DNA using some stress factors such as heat or electrical shock.
- **Microinjection-** a sharp needle is used for insertion of DNA directly into the nucleus of a cell, however, the method is less effective and required a higher level of expertise for that.
- **Electroporation-** one of the best methods having a great success rate. Recombinant DNA is inserted into the host genome by permeabilizing the cell with electrical current.
- **Sonication-** sonication is yet another good method sometimes used in the gene transfer experiment in which the recombinant DNA is inserted into the target cell using ultrasonic waves. The ultrasonic waves also increase the permeability of cells.
- **Liposome mediated gene transfer-** Using an artificial cell-like outer coat known as a liposome- recombinant DNA can be inserted in the host genome.
- **Gene transfer using bacterial infection-** This method is one of the popular methods and routinely used in plant genetic engineering experiments. Here, the plant species is infected with the transformed bacteria for inserting a gene of interest.

Confirmation of insert

- In the traditional culturing method, the presence or absence of a selectable marker is used to differentiate transformed cells from the untransformed cells.
- **PCR based detection method** - widely accepted & trusted method. DNA is extracted from the transformed cell and amplified using the primers complementary to our gene of interest or our recombinant DNA. If the recombinant DNA is present it surely amplified otherwise no amplification obtained. For the two factor conformation, one primer set complementary to recombinant DNA specific and one set of primer complementary to the selectable marker sequence are taken and multiplex PCR is performed. For validating results, amplification must be obtained in both the reaction.
- **What happened if any mutation occurred** during the experiment in our gene of interest? Because the PCR can only amplify the DNA. We must need sequence information to detect the mutation. For that, the DNA sequencing method is used.
- DNA is extracted from the transformed cells and the gene of interest is amplified using the PCR. Now the PCR amplicons are used for DNA sequencing in which using the fluorescent chemistry the sequence of our gene of interest is orderly determined.
- Once all the parameters for determining the gene of interest fulfilled, our cells are now ready to inject in the host organism or for tissue culture experiments.

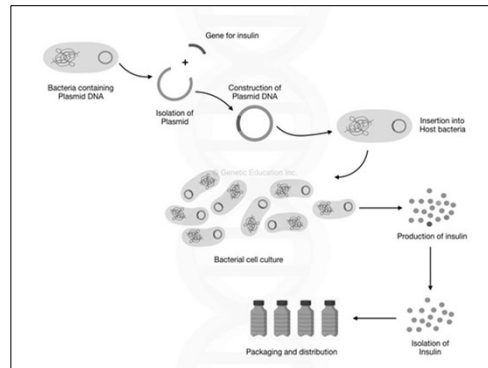
Applications of r-DNA Technology

- rDNA technology has great industrial and agricultural value. It is practiced in medicine, genetic research, agriculture, crop improvement, and for production of therapeutic drugs. It is also used in the development of genetically modified organisms.
- The recombinant DNA technology is used in the crop improvement and development of new economically important traits -
 - ❖ Herbicide resistance
 - ❖ Virus resistance
 - ❖ Delayed fruit ripening
 - ❖ Altered oil content
 - ❖ Pollen control
 - ❖ Development of cold and drought-tolerant plant species.
 - ❖ A classical example of it is the **BT cotton**- one of the types of genetically modified species provides resistance to the plant against *Bacillus thuringiensis* and many types of moths and butterflies can be controlled.

Applications in Medical

- Low-cost drugs, hormones, enzymes, and vaccines are created using genetic engineering tools e.g. -
- **Anti-blood-clotting factor** is the best example of it in which the plasminogen activating enzyme which is capable of dissolving the blood clot is artificially designed and used in the patients with **coronary artery disease** or **heart attack**.

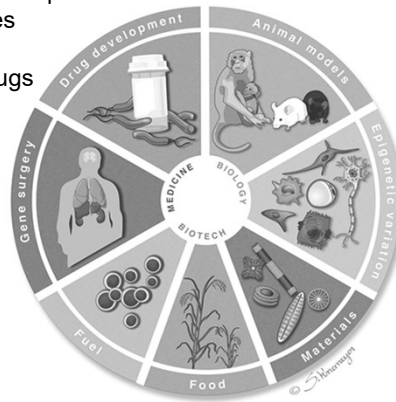
- Therapeutic proteins **somatostatin** and **lymphokines** which are worked against several disease conditions and can be synthesized artificially.
- **Insulin** is yet a classic example of a therapeutic protein designed using genetic engineering technology.
- Recombinant vaccines against smallpox, herpes simplex virus and hepatitis.
- Created a unique vaccines that only contains the DNA for viral coat protein thus the pathogen can never be activated again.



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Applications

- The purpose of developing the genetic engineering or genetic manipulating technique is to **produce organisms or phenotypes which are useful to us**. Genetic engineering techniques are used for -
 - ❖ Construction of genetically modified plant species
 - ❖ Abiotic and biotic stress resistant plant species
 - ❖ Economically important plant species
 - ❖ Commercially valuable organism
 - ❖ For the production of therapeutic drugs
 - ❖ Prevention of genetic abnormalities



Limitations

- There are ethical issues associated with the use of gene therapy and genetically engineered products.
- Also, to provide an economic value to the food product or any GM product, the nutritional values are compromised.
- Because of the adverse effect of it, new resistant pathogenic strains are evolved faster. Also, the side effects of gene therapy and the use of viruses in it may sometimes harmful to the target organism.
- The technology is not so cost effective.
- Playing with the embryo or fetus is against the natural law, people strongly believe in it, thus genetically modified food and plant products are always becoming a center of controversy.
- **Overall, positive use of rDNA technology can change the fate of mankind.**

THANK YOU